

Chuan Zhang

M.Sc. Student in Mathematics, University of Victoria

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RESEARCH INTERESTS

Computational complexity; quantum algorithms; graph algorithms; combinatorics; discrete optimization.

EDUCATION

University of Victoria, Victoria, BC

Sep 2025–Aug 2027 (expected)

M.Sc. in Mathematics | GPA: 9.00/9.00

Supervisor: Jing Huang

Thesis working title: *On Structural Aspects of Strongly Chordal Digraphs*

Simon Fraser University, Burnaby, BC

2022

B.Sc. in Computing Science, Minor in Mathematics | CGPA: 3.44/4.33

Shanghai Jiao Tong University, Shanghai, China

Apr 2016

B.Eng. in Mechanical Engineering

RESEARCH MANUSCRIPTS

Strongly Chordal Semicomplete Digraphs

Complete draft; not submitted

Jing Huang and Chuan Zhang

- Developed forbidden-subdigraph and structural characterizations for the irreflexive case.
- Designed a polynomial-time recognition algorithm for the irreflexive case.
- Developed the reflexive construction independently; the supervisor later identified it as an adjusted-interval-graph construction, and its recognition procedure reuses the irreflexive algorithm.

Strong Chordality of Locally Semicomplete Digraphs

Manuscript in preparation

Jing Huang and Chuan Zhang

- Constructed and proved strong orderings for straight and round local-tournament cases.
- Extended these constructions to admissible tournament blowups using the supervisor's framework.
- Extending the analysis to reduced multi-component cases and writing the manuscript's main text.

ONGOING RESEARCH

Strong Orderings in Transitive Oriented Graphs

With Jing Huang

- Investigating minimal-obstruction characterizations for strong orderings in transitive oriented graphs.
- Exploring infinite-family or classification-based descriptions of the obstructions.

RELEVANT COURSEWORK

Selected graduate coursework: Quantum Algorithms and Software Engineering (100%); Computational Complexity (99%); Combinatorial Designs (98%); Algebraic Methods in Combinatorics and Discrete Optimization (95% each).

Additional theory coursework: Randomized Algorithms, Cryptography, Graph Theory, Computational Algorithms, Algorithms of Optimization, Introduction to Quantum Computing, and Combinatorial Theory.

TEACHING EXPERIENCE

Teaching Assistant, CSC 226: Algorithms and Data Structures II **Fall 2025**
University of Victoria

- Co-developed laboratory materials and facilitated laboratory instruction.
- Delivered one guest lecture.

Marker, Department of Mathematics and Statistics **2025–2026**
University of Victoria

- MATH 102 (Fall 2025); MATH 151, MATH 211, and MATH 322 (Spring 2026).

AWARDS AND HONOURS

Betty and Gilbert Kennedy Graduate Scholarship in Mathematics and Statistics **2025**
University of Victoria | Sole recipient

Albert Hung Chao Hong Scholarship **2019**
University of Victoria | Sole recipient

TECHNICAL SKILLS

Programming: Python, C++ **Tools:** $\text{\LaTeX}$, Git

Quantum software: Experience with PennyLane and Qiskit